**Title:** Does removing race from eGFR change who an AI misses? Fairness of machine-learning screening for undiagnosed chronic kidney disease in NHANES 2017–March 2020

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**Background & Objective:** About 9 in 10 U.S. adults with chronic kidney disease (CKD) are unaware of it. We developed a two-stage machine learning screening framework for primary care and audited whether replacing race-adjusted CKD-EPI 2009 with race-free CKD-EPI 2021 changes CKD classification and missed cases.

**Methods:** NHANES 2017–March 2020 adults ≥20 years without prior kidney disease diagnosis (N = 7,426). Undiagnosed CKD: eGFR <60 mL/min/1.73 m² (both equations) or UACR ≥30 mg/g. Stage 1 (EHR Zero-Lab) used non-invasive sociodemographic and vital variables; Stage 2 added glucose, uric acid and cholesterol (creatinine, UACR and urea nitrogen excluded to prevent leakage). Race was not a predictor and was used only for auditing. Elastic-net logistic regression and XGBoost were trained on CKD-EPI 2021 (70/30 split, survey-weighted) and evaluated against both equations; the threshold was prespecified for ≥90% out-of-fold sensitivity. The fairness audit compared false-negative rate (FNR) and specificity in low-income (PIR <1.35) non-Hispanic Black participants versus others (Korn–Graubard and MOVER intervals).

**Results:** Weighted undiagnosed CKD prevalence was 12.5% (11.5–13.6) with CKD-EPI 2021 and 13.5% (12.3–14.9) with 2009; 130 participants (1.75%) were reclassified, including 14/599 low-income Black participants, all becoming cases under 2021. In the test set (n = 2,229), weighted AUC was 0.736 (0.690–0.783) for Stage 1 XGBoost, 0.757 (0.714–0.801) for Stage 2, and 0.769 (0.725–0.814) for logistic regression (ΔAUC −0.033; −0.061 to −0.005). At threshold 0.0221, sensitivity was 91.2%, specificity 22.5%, and 79.2% were flagged. Under CKD-EPI 2021, FNR was 0% (0/31; 0–11.2) in the subgroup versus 9.4% (5.2–15.3) in others (difference −9.4 points; −15.3 to 2.6); under 2009, 0% (0/28; 0–12.3) versus 8.3% (4.7–13.3). Specificity was 14.9% (5.6–29.7) versus 22.9% (19.6–26.5) (difference −8.1; −18.0 to 7.2).

**Limitations & Conclusions**: Limitations: cross-sectional design, eGFR-based definition, only 31 subgroup cases, no external validation. Switching equations reclassified few participants and did not change whom the model missed: no subgroup cases were missed, but 87% of the subgroup was flagged. Overall, the model flagged most participants and underperformed logistic regression, limiting its triage value.